

Naive Bayes Foundations: Mutually Exclusive Events

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Leads to:** Naive Bayes ([Note 87](#)); Voting ensembles ([Note 102](#)).

1. Overview

Key point: Two events are mutually exclusive when they cannot happen together: $P(A \cap B) = 0$. This is a different idea from independence, and the two are easily confused.

The previous Note defined **independent** events: they can happen together, but one does not affect the other. **Mutually exclusive** events are different: they can never happen together. The two ideas are easy to mix up, so this Note puts them side by side.

2. The definition

Key point: Mutually exclusive: $P(A \cap B) = 0$, so $P(A | B) = 0$.

Two events A and B are **mutually exclusive** if

$$P(A \cap B) = 0$$

They have no outcome in common, so at most one of them can happen.

Examples:

- A single coin toss cannot land both heads and tails.
- At a junction, a driver turns either left or right, not both.
- One die cannot show both 3 and 6 on the same roll.

3. Conditional probability for mutually exclusive events

Key point: If B happened, A cannot have happened: $P(A | B) = 0$.

Putting $P(A \cap B) = 0$ into the conditional probability formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0$$

With numbers: die 1 has shown 3. The probability that the same die shows 6 on that roll is 0. A driver has turned left; the probability that they turned right is 0.

4. Mutually exclusive is not independent

Key point: Knowing B happened tells us A did not, which is a lot of information. So mutually exclusive events (each with a positive probability) are always dependent.

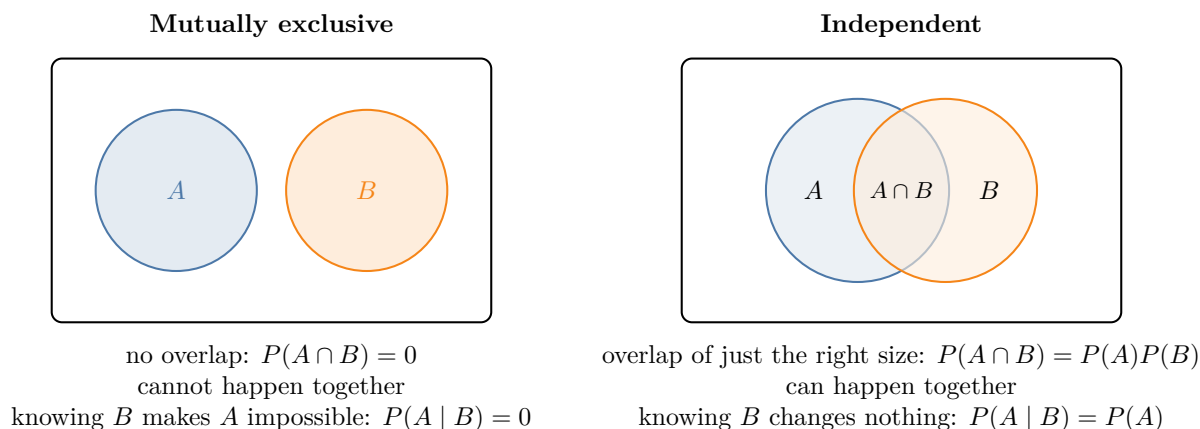


Figure 1: Mutually exclusive versus independent events

	Mutually exclusive	Independent
Can happen together?	no	yes
$P(A \cap B)$	0	$P(A) \times P(B)$
$P(A B)$	0	$P(A)$
Does B tell us about A?	yes: A did not happen	no

For “die 1 shows 3” (A) and “die 1 shows 6” (B), each has probability 1/6:

- $P(A \cap B) = 0$, so they are mutually exclusive.
- $P(A) \times P(B) = 1/36 \neq 0$, so they are **not** independent.

In fact this always happens: if both events have a probability above 0, being mutually exclusive rules out being independent. Learning that one happened tells us for certain that the other did not.

Extra: For mutually exclusive events, the probability that one **or** the other happens is a simple sum: $P(A \cup B) = P(A) + P(B)$. Here $1/6 + 1/6 = 1/3$. This “addition rule” is used in the Bayes’ theorem Notes to split a probability into separate cases.

5. Summary

- Mutually exclusive: $P(A \cap B) = 0$; the events cannot happen together, and $P(A | B) = 0$.
- Independent: $P(A \cap B) = P(A)P(B)$; they can happen together, and $P(A | B) = P(A)$.
- Mutually exclusive events with positive probabilities are never independent.

6. Key terms

Term	Meaning
Mutually exclusive events	Events that cannot happen at the same time; their intersection has probability 0
Union ($A \cup B$)	The event that A or B (or both) happens
Addition rule	For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$